

LRSC Odd-Ring Spectral Rank Theorem

Version 1.2 - Analytic Family Result

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Abstract

For a single-step near-Nyquist local redistribution map on an odd cyclic lattice, this note proves that the rank of the conjugate-balanced post-stencil channel family equals the number of active reflection orbits, provided at least one reflection orbit is inactive. The proof closes the previously isolated non-vanishing condition for the Fourier coefficients of the local shedding field for every odd $M \geq 9$, using residue-class trigonometric formulas. A Chebyshev/Vandermonde argument gives the pre-stencil rank, and injectivity of the stencil on the masked channel span preserves it. The result is structural and does not imply a universal fixed-unit compression number K_τ .

1 Setup and theorem

Let $M \geq 9$ be odd and

$$k = \frac{M-1}{2}, \quad \alpha = \frac{\pi}{M}.$$

On $\mathbb{Z}_M$, define the near-Nyquist carrier

$$X_t = 1 + A \cos\left(\frac{2\pi kt}{M}\right), \quad A > 0,$$

and the local shedding field

$$j_t = X_t - \min\{X_{t-1}, X_t, X_{t+1}\}.$$

Let $m_t \in \{0, 1\}$ be any reflection-symmetric mask, $m_t = m_{-t}$, with at least one inactive reflection orbit. Let p denote the number of active reflection orbits. Form the conjugate-balanced pre-stencil channel family generated by the Fourier coefficients of j , and let S be the circulant stencil whose Fourier multiplier is

$$s_q = \cos\left(\frac{2\pi q}{M}\right) - 1.$$

Theorem. Under these assumptions, the post-stencil channel matrix $V = SW$ satisfies

$\text{rank}(V) = p.$

2 Exact shedding field

Using centered representatives $t = -k, \dots, k$,

$$\cos\left(\frac{2\pi kt}{M}\right) = \cos((\pi - \alpha)t) = (-1)^t \cos(\alpha t),$$

so

$$X_t = 1 + A(-1)^t \cos(\alpha t).$$

For odd $|t|$, X_t is the local minimum and $j_t = 0$. At the origin,

$$j_0 = A(1 + \cos \alpha).$$

For even $|t| > 0$, monotonicity of cosine on the centered half-ring gives

$$\boxed{j_t = A [\cos(|t|\alpha) + \cos((|t| - 1)\alpha)]}.$$

Thus j is real and reflection-even.

Let

$$L = \left\lfloor \frac{M - 1}{4} \right\rfloor.$$

Then

$$\hat{j}_r = \frac{1}{M} \left[j_0 + 2 \sum_{u=1}^L j_{2u} \cos\left(\frac{4\pi ru}{M}\right) \right].$$

Finite trigonometric summation reduces this expression as follows.

3 Non-vanishing of the shedding spectrum

3.1 $M = 4L + 3$

For $0 \leq r < k$,

$$\hat{j}_r = \frac{2A \cos(\alpha/2)}{M} B_r,$$

where

$$B_r = -\frac{\sin(2\alpha) [(-1)^r \cos(r\alpha) + \sin(\alpha/2)]}{2 \sin((2r - 1)\alpha) \sin((2r + 1)\alpha)}.$$

The denominator is nonzero for integer $0 \leq r < k$. For even r , the bracket is positive. For odd r ,

$$r\alpha \leq (k - 1)\alpha = \frac{\pi}{2} - \frac{3\alpha}{2},$$

hence

$$\cos(r\alpha) \geq \sin(3\alpha/2) > \sin(\alpha/2).$$

Thus the bracket is strictly negative. Therefore

$$\boxed{\hat{j}_r \neq 0 \quad (0 \leq r < k)}$$

for every $M \equiv 3 \pmod{4}$.

3.2 $M = 4L + 1$

Here $k = 2L$ and

$$\widehat{j}_r = \frac{2A \cos(\alpha/2)}{M} B_r,$$

with

$$B_r = -\frac{\sin \alpha [(-1)^r \cos(r\alpha) \cos(2r\alpha) + \cos \alpha \sin(\alpha/2)]}{\sin((2r-1)\alpha) \sin((2r+1)\alpha)}.$$

Again the denominator is nonzero for $0 \leq r < k$. A zero would require

$$|\cos(r\alpha) \cos(2r\alpha)| = c, \quad c := \cos \alpha \sin(\alpha/2).$$

For $0 \leq r \leq L-1$, $f(x) = \cos x \cos 2x$ is strictly decreasing on $[0, \pi/4]$. Hence

$$f(r\alpha) \geq f((L-1)\alpha) = \cos\left(\frac{\pi}{4} - \frac{5\alpha}{4}\right) \sin\left(\frac{5\alpha}{2}\right).$$

Since $M \geq 9$, $\alpha \leq \pi/9$. Therefore

$$\cos\left(\frac{\pi}{4} - \frac{5\alpha}{4}\right) > \frac{1}{\sqrt{2}}, \quad \sin\left(\frac{5\alpha}{2}\right) \geq \frac{5\alpha}{\pi},$$

while $c < \alpha/2$. Because $5/(\pi\sqrt{2}) > 1/2$, one has $f(r\alpha) > c$.

At $r = L$,

$$|\cos(L\alpha) \cos(2L\alpha)| = \cos(L\alpha) \sin(\alpha/2).$$

Since $L\alpha = \pi/4 - \alpha/4 > \alpha$, $\cos(L\alpha) < \cos \alpha$, so this quantity is strictly less than c .

For $L+1 \leq r \leq 2L-1$, put $s = 2L - r$ and $t = (s+1/2)\alpha$. Then

$$|\cos(r\alpha) \cos(2r\alpha)| = h(t) := \sin t \cos 2t,$$

where

$$\frac{3\alpha}{2} \leq t \leq \frac{\pi}{4} - \frac{3\alpha}{4}.$$

Moreover

$$h'(t) = \cos t (1 - 6 \sin^2 t),$$

so h has at most one critical point in this interval, and that point is a maximum. The minimum is therefore attained at an endpoint.

At $t = 3\alpha/2$,

$$\frac{h(3\alpha/2)}{c} = \frac{\sin(3\alpha/2) \cos 3\alpha}{\sin(\alpha/2) \cos \alpha} = \frac{(1 + 2 \cos \alpha) \cos 3\alpha}{\cos \alpha} > 1,$$

because $\alpha \leq \pi/9$ implies $\cos 3\alpha \geq 1/2$. At the other endpoint,

$$\frac{h(\pi/4 - 3\alpha/4)}{c} = \frac{\sin(\pi/4 - 3\alpha/4)}{1} \frac{1 + 2 \cos \alpha}{\cos \alpha} > 1,$$

because $\sin(\pi/4 - 3\alpha/4) \geq 1/2$.

Hence equality with c is impossible for every integer $0 \leq r < k$, and

$$\boxed{\widehat{j}_r \neq 0 \quad (0 \leq r < k)}$$

for every $M \equiv 1 \pmod{4}$ as well.

Combining the two residue classes proves the non-vanishing lemma for every odd $M \geq 9$.

4 Pre-stencil rank

Since j is real and reflection-even, the conjugate-balanced channels have the spatial form

$$w_0(i) = \hat{j}_0 m_i,$$

and for $r \geq 1$,

$$w_r(i) = 2\hat{j}_r m_i \cos\left(\frac{2\pi r i}{M}\right).$$

An odd ring has $k + 1$ reflection orbits. At least one is inactive, so $p \leq k$. The non-vanishing lemma therefore applies to the first p channel scales.

Choose one representative i_s from each active reflection orbit and set

$$x_s = \cos\left(\frac{2\pi i_s}{M}\right).$$

The x_s are distinct. After removal of nonzero row and column scales, the resulting $p \times p$ matrix is

$$C_{rs} = T_r(x_s), \quad 0 \leq r, s \leq p-1,$$

where T_r is the Chebyshev polynomial of degree r . Its determinant is a nonzero constant multiple of

$$\prod_{0 \leq i < j < p} (x_j - x_i),$$

so the first p channels are linearly independent. Reflection symmetry bounds the active masked space by dimension p , hence

$$\boxed{\text{rank}(W) = p.}$$

5 The stencil preserves rank

The multiplier

$$s_q = \cos\left(\frac{2\pi q}{M}\right) - 1$$

vanishes only at $q = 0$, so the spatial kernel of S consists exactly of constant vectors. Every vector in $\text{span}(W)$ vanishes on every inactive site because each channel contains the factor m_i . Therefore the only constant vector in $\text{span}(W)$ is zero:

$$\ker S \cap \text{span}(W) = \{0\}.$$

Thus S is injective on the channel span and

$$\boxed{\text{rank}(V) = \text{rank}(SW) = \text{rank}(W) = p.}$$

This proves the theorem.

6 Locked benchmark and computational audit

For the locked $M = 99$, $A = 0.50$, $\theta = 0.08$ benchmark, there are 41 active reflection orbits; the theorem gives $\text{rank}(V) = 41$. Under the same A, θ construction, the tested neighboring rings give 42 active reflection orbits for $M = 101$ and 52 for $M = 127$, hence ranks 42 and 52 respectively.

The companion verifier checks the closed shedding field, compares the simplified Fourier formulas with direct FFT coefficients for all odd $M = 9, \dots, 2001$, and performs exact integer/cyclotomic non-vanishing checks for $M = 99, 101, 127$. These calculations corroborate the analytic proof but do not replace it.

7 Claim firewall

This theorem concerns the rank of a single-step phenomenological cyclic redistribution operator. It does not establish a microscopic physical law, a universal compression dimension, a universal value of K_τ , a multi-step dynamical theorem, or a result about the separate entropy-to-horizon bridge question.